File: SIDE-PROXIMITY-IR-SENSOR .kicad_sch

SIDE PROXI SENSOR

IR-L-RX
CE-L
GPIO1-L
IR-R-RX
CE-R
GPIO1-R
SDA
SCL

CLIFF

RR-Q+
RF-Q+
LR-Q+
LF-Q+
BUMP-R-Q+
BUMP-L-Q+
BUMPER_LED_EN
CLIFF_LED_EN

File: CLIFF.kicad_sch

File: WATER-PUMP.kicad_sch

WATER PUMP

WATER-PUMPU-CTRL
WATER-PUMP-SENSE-ADC

MAIN BRUSH MOTOR

MAIN-BRUSH-IN1
MAIN-BRUSH-IN2
MAIN-BRUSH-CURRENT-ADC

File: MAIN-BRUSH-MOTOR .kicad_sch

File: WHEEL-MOTORs .kicad_sch

WHEEL MOTORs

WHEEL-M-RIGHT-IN2
WHEEL-M-RIGHT-IN1
WHEEL-M-RIGHT-ENCODE-AD
WHEEL-M-RIGHT-DROPD
WHEEL-M-LEFT-IN2
WHEEL-M-LEFT-IN1
WHEEL-M-LEFT-DROPD
WHEEL-M-LEFT-ENCODE-AD
WHEEL-M-R-F-ADCD
WHEEL-M-L-F-ADCD

MCU-STM32

WATER-PUMPU-CTRL
WATER-PUMP-SENSE-ADC
MAIN-BRUSH-IN1
MAIN-BRUSH-IN2
MAIN-BRUSH-CURRENT-ADC
WHEEL-M-RIGHT-IN2
WHEEL-M-RIGHT-IN1
WHEEL-M-RIGHT-ENCODE-A
WHEEL-M-RIGHT-DROP
WHEEL-M-LEFT-IN2
WHEEL-M-LEFT-IN1
WHEEL-M-LEFT-DROP
WHEEL-M-LEFT-ENCODE-A
WHEEL-M-R-F-ADC
WHEEL-M-L-F-ADC
MAIN-FAN-S-SENSE
MAIN-FAN-S-CTRL
SIDE-BRUSH-IN1
SIDE-BRUSH-IN2
SIDE-BRUSH-CURRENT-ADC
I2C2_SDA
I2C2_SCL
IMU-INT#2
IMU-INT#1
IMU-FSYNC
LED-HOME
LED-PWR
CLEAN-BTN
HOME-BTN
USB_PWR_EN
DM
DP
BAT_ID
5V_EN
3.3V_EN
PWR-BTN-SENSE
STM-PWR-CTRL
PI_SHUTDOWN
PI_DONE
STM32-UART1-RX
STM32-UART1-TX
PI-RESET
PMIC_PWROND
CLIFF_LED_EN
BUMPER_LED_EN
BUMP-L-Q+
BUMP-R-Q+
LF-Q+
LR-Q+
RF-Q+
RR-Q+
USART4-IR-R-RX
SIDE-PROXI-CE-L
SIDE-PROXI-GPIO-L
USART2-IR-L-RX
SIDE-PROXI-CE-R
SIDE-PROXI-GPIO-R
I2C4_SDA
I2C4_SCL
CARPET-SENSOR-ECHO
VM-VBAT-END
CARPET-SENSOR-LOD
STM32_RST
WDI
LIDAR_RXD
LIDAR_TXD
LIDAR-M-CTRL
LIDAR_EN
UART3_RX2
UART3_RX1
LIGHT-ON
BOOT
SWCLK
SWDIO
STM_RST
PI_SHUTDOWN
PI_DONE
UART2_TX
UART2_RX
PMIC_EN
RUN_PG
IMU-INT2
IMU-INT1
IMU-FSYNC
VM-VBAT-EN
STM-PWR-CTRL
PWR-BTN-SENSE
3.3V_EN
5V_EN
POWER-BTN
M2_PWR_EN
PCIE_PWR_EN

File: STM32G473.kicad_sch

MAIN FAN

MAIN-FAN-S-SENSED
MAIN-FAN-S-CTRLG

File: MAIN-FAN .kicad_sch

SIDE BRUSH MOTORs

SIDE-BRUSH-IN1
SIDE-BRUSH-IN2
SIDE-BRUSH-CURRENT-ADCD

File: SIDE-BRUSH-MOTORs .kicad_sch

File: IMU-ICM-4267-P .kicad_sch

IMU

IMU-SDA
IMU-SCL
IMU-INT#2D
IMU-INT#1D
IMU-FSYNC

BUTTON-LEDs

LED-HOME
LED-PWR
CLEAN-BTND
HOME-BTND
POWER-BTND

File: BUTTON-LEDs .kicad_sch

POWER CONTROL

STM-PWR-CTRL
PWR-BTN-SENSE
3.3V_EN
5V_EN
POWER-BTN
VM-VBAT-EN
M2_PWR_EN

File: SYSTEM-POWER.kicad_sch

BATTERY-CHARGER

BATTERY CHARGER

ID
D+
D-
PWR_EN

File: Battery-Charger.kicad_sch

CM5-GPIO

BOOT0
SWCLK
SWDIO
STM_RST
PI_SHUTDOWN
PI_DONE
UART2_TX
UART2_RX
PMIC_EN
RUN_PG
GPIO25
GPIO26
GPIO27
GPIO16
GPIO17
SCL1
SDA1
GPIO06/GPCLK2
I2S_SCLK
I2S_LRCLK
I2S_DOUT
I2S_DIN
GPIO011
MIPI1_IO0
MIPI1_IO1
CAM_GPIO0
CAM_GPIO1
SCL0
SDA0
ID_SC
ID_SD

CM5-GPIOS

File: CM5-GPIO.kicad_sch

CM5-HIGHSPEED

MIPI0_D0_P
MIPI0_D0_N
MIPI0_D1_P
MIPI0_D1_N
MIPI0_C_P
MIPI0_C_N
MIPI1_D0_P
MIPI1_D0_N
MIPI1_D1_P
MIPI1_D1_N
MIPI1_C_P
MIPI1_C_N
CM4_CAM0_D0_P
CM4_CAM0_D0_N
CM4_CAM0_D1_P
CM4_CAM0_D1_N
CM4_CAM0_C_P
CM4_CAM0_C_N
PCIE_CLK_nREQ
PCIE_nRST
PCIE_nWAKE
PCIE_TX_P
PCIE_TX_N
PCIE_RX_P
PCIE_RX_N
PCIE_CLK_P
PCIE_CLK_N

File: CM5-Highspeed.kicad_sch

CARPET-SENSOR

CARPET-SENSOR

ECHO_ADC
LO

File: Carpet-sensor.kicad_sch

WATCHDOG

WD

WDI

File: WATCHDOG.kicad_sch

LIDAR

LIDAR-RXD
LIDAR-TXD
LIDAR-MOTOR-CTRL
LIDAR-EN

File: LIDAR.kicad_sch

FRONT SENSORS

IR-F-RX2
IR-F-RX1
LIGHT-ON
VL-L-INT
VL-L-LPn
VL-R-INT
VL-R-LPn
VL-I2C-RESET
VL-I2C-SCL
VL-I2C-SDA

File: Front-Sensors.kicad_sch

SPEAKER

BCLK
LRCLK
DIN
SP_SHUTDOWN

File: SPEAKER.kicad_sch

CAMERA

MIPI1_IO0
MIPI1_IO1
CAM_GPIO0
CAM_GPIO1
SCL0
SDA0
ID_SC
ID_SD
MIPI0_D0_P
MIPI0_D0_N
MIPI0_D1_P
MIPI0_D1_N
MIPI0_C_P
MIPI0_C_N
MIPI1_D0_P
MIPI1_D0_N
MIPI1_D1_P
MIPI1_D1_N
MIPI1_C_P
MIPI1_C_N
CM4_CAM0_D0_P
CM4_CAM0_D0_N
CM4_CAM0_D1_P
CM4_CAM0_D1_N
CM4_CAM0_C_P
CM4_CAM0_C_N

File: MIPI-CAMERA.kicad_sch

M.2

M.2 CONN

PCIE_CLK_nREQ
PCIE_nRST
PCIE_nWAKE
PCIE_TX_P
PCIE_TX_N
PCIE_RX_P
PCIE_RX_N
PCIE_CLK_P
PCIE_CLK_N

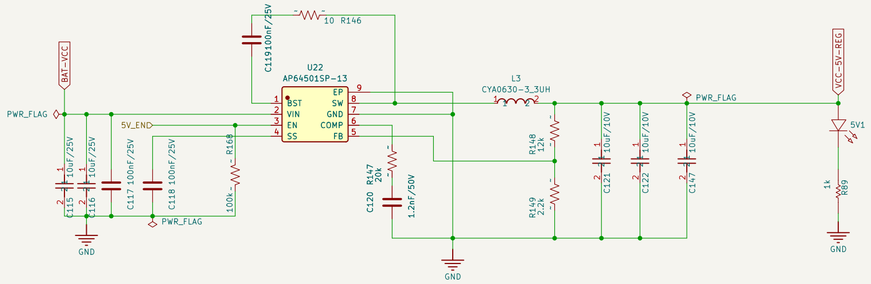
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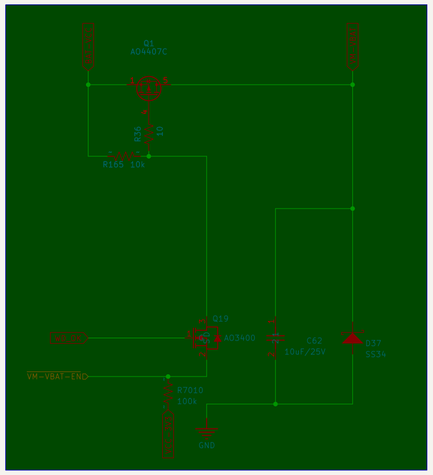
Sheet: /
File: Main.kicad_sch

Title:

Size: A2 | Date: | Rev:
KiCad E.D.A. 10.0.4 | Id: 1/22

POWER MANAGMENT

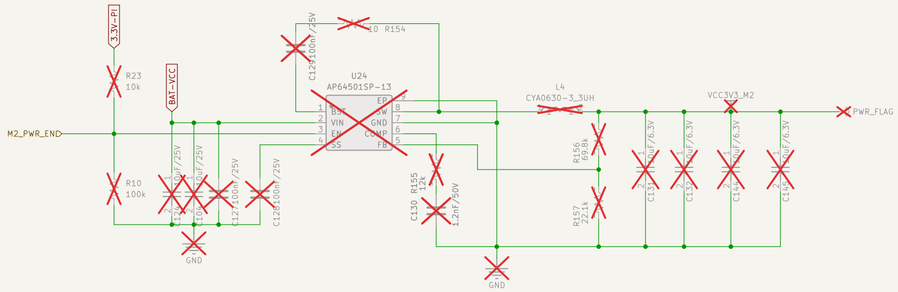


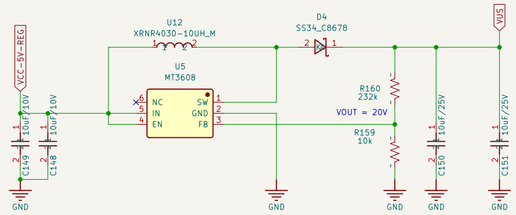


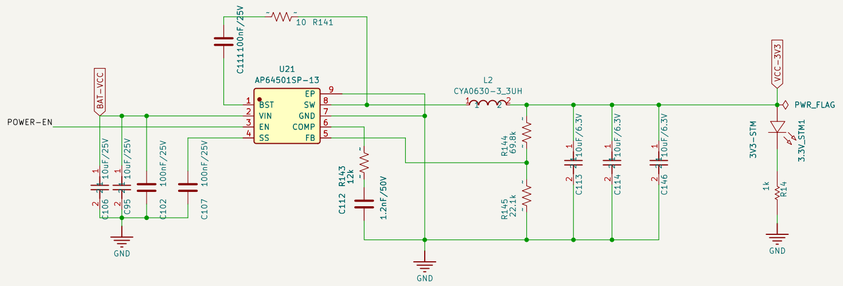
TODO Check VM-VBAT max current vs Q1 rating

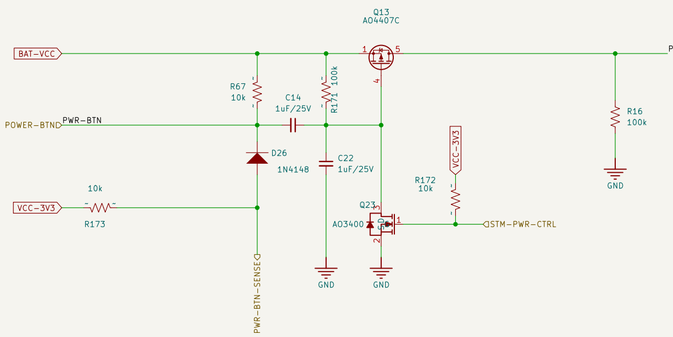
Old VM-5V switch removed: the LiDAR (VM-5V-LIDAR) and the water pump (VM-5V-WATER-PUMP) now have their own WD_OK-gated switches on their sheets.
No soft ramp.

VM-VBAT (motors, fan) is on only while WD_OK is high (RC watchdog fed) AND VM-VBAT-EN is low.
Q19 source-switched AND: GPIO low = on. R7010 keeps it off in reset/flash. AO4407C: Vgs +/-25 V, no gate divider needed.

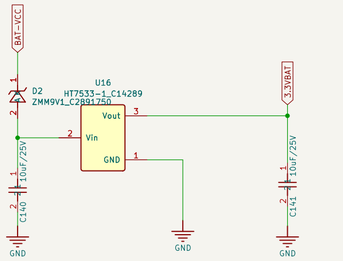




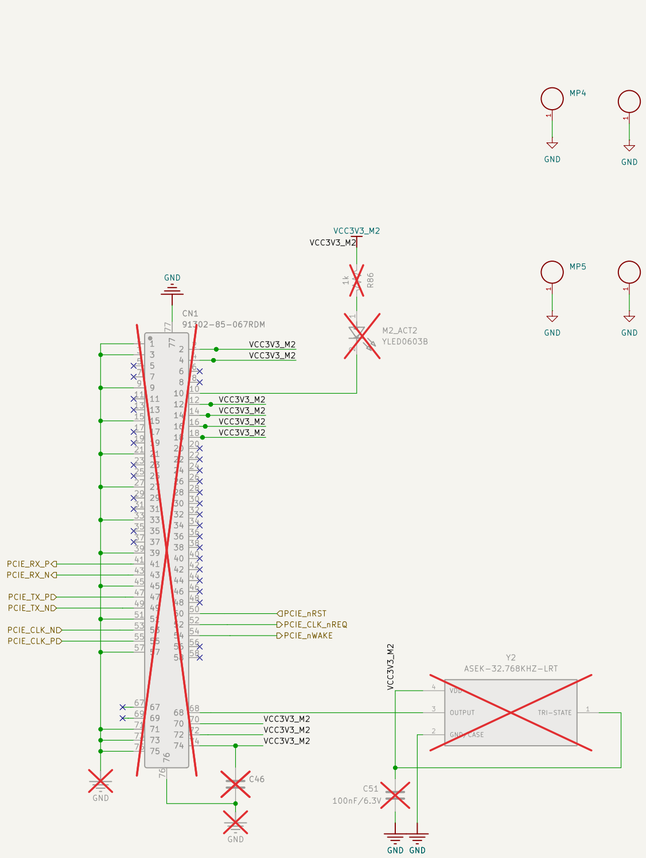




STM-PWR-CTRL need to use for both EN and OFF from STM32 GPIO pin.
1- Hold EN to HIGH after buttone pressed
2- OFF board

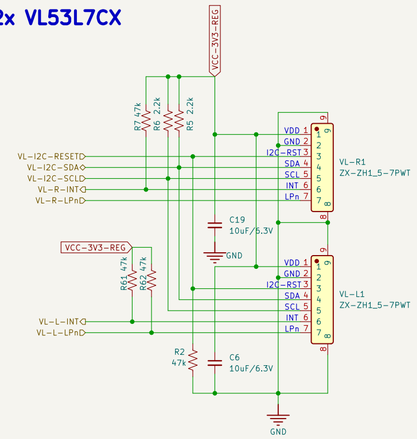


Sheet: /POWER/		
File: SYSTEM-POWER.kicad_sch		
Title:		
Size: A2	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 2/22

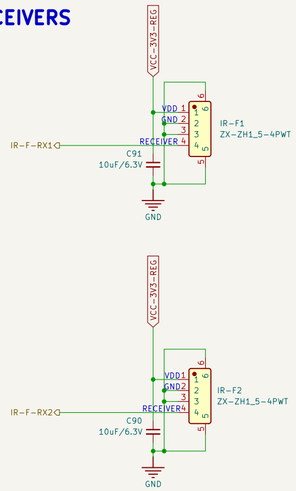


Sheet: /M.2/	
File: M.2.kicad_sch	
Title:	
Size: A2	Date:
KiCad E.D.A. 10.0.4	**Rev:**
	Id: 3/22

2x VL53L7CX

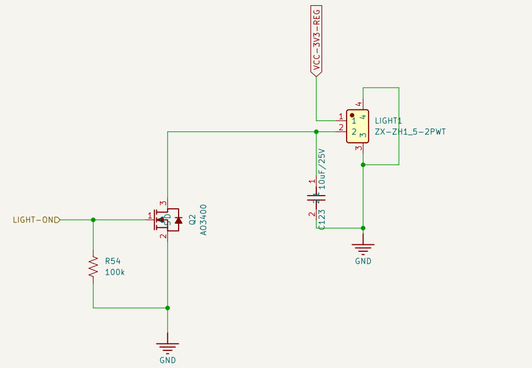


2 IR RECEIVERS



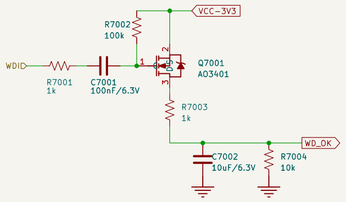
IR LIGHT

https://thepihut.com/products/850nm-ir-led-modules?srsltid=AfmBOorZOhl4HyIQHd5JbNfaA7bqJMbzrmYObn7Mm12pD_mKEoIREOnX



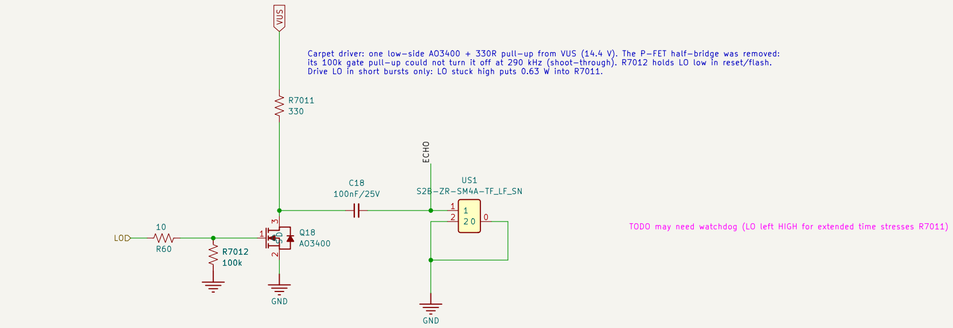
Sheet: /FRONT SENSORS/		
File: Front-Sensors.kicad_sch		
Title:		
Size: A2	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 4/22

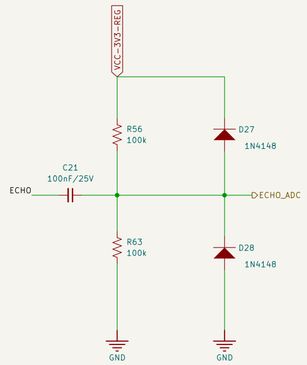
RC CHARGE-PUMP WATCHDOG: WD_OK gates VM-VBAT, VM-5V-LIDAR and VM-5V-WATER-PUMP



Firmware toggles WDI in software (50-1000 Hz, ~50% duty) only while the control loop is healthy.
Never drive WDI from a timer/PWM: a timer keeps running after the CPU hangs.
Each falling edge on WDI pulls the Q7001 gate low through C7001 and tops up C7002 via R7003.
WD_OK ~2.7 V while toggling. WDI stuck high, low or floating: <1 V in ~100 ms, <0.4 V in ~190 ms
(10 uF nominal; DC-bias derating shortens this). WD_OK starts at 0 V at power-up (fail-safe).
WD_OK drives N-FET gates only (14.4 V rail switch, water pump, LIDAR incl. laser): no resistive loads.
WDI pin: high-Z at reset, no pull-down. All parts reuse existing BOM lines.

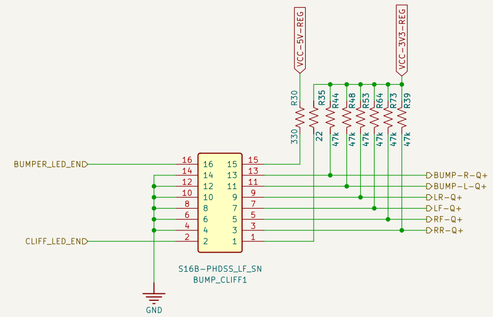
Sheet: /WATCHDOG/		
File: WATCHDOG.kicad_sch		
Title:		
Size: A2	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 5/22





Sheet: /CARPET-SENSOR/	
File: Carpet-sensor.kicad_sch	
Title:	
Size: A2	Date:
KiCad E.D.A. 10.0.4	**Rev:** Id: 6/22

ANTI FALL SENSORs

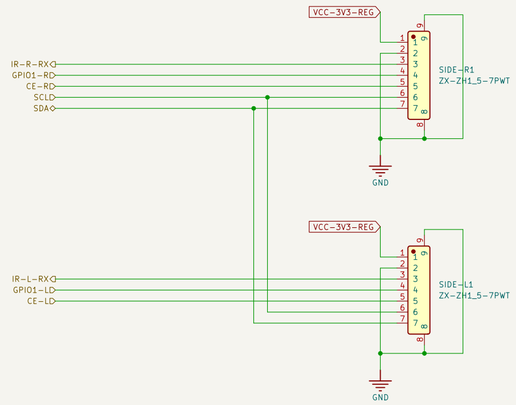


Sheet: /CLIFF/
File: CLIFF.kicad_sch

Title:

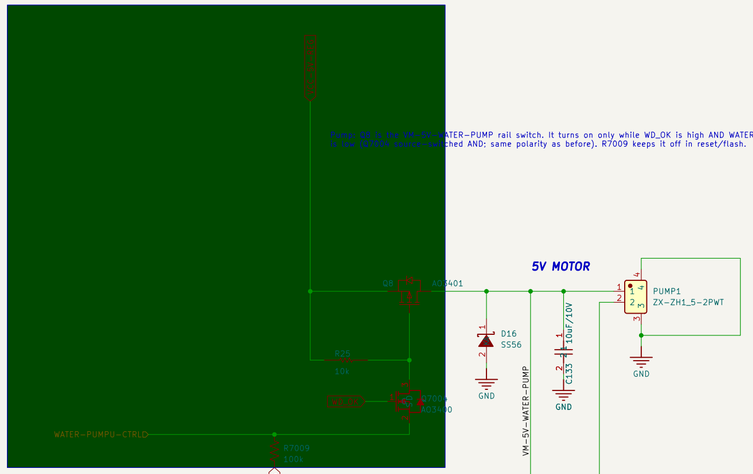
Size: A2	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 19/22

SIDE PROXI IR SENSORs



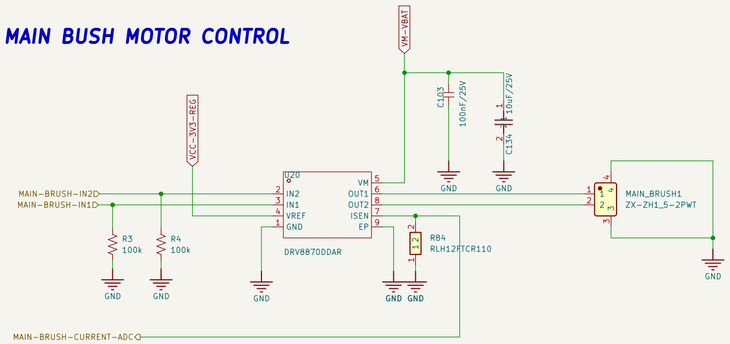
Sheet: /SIDE-PROXI/		
File: SIDE-PROXIMITY-IR-SENSOR .kicad_sch		
Title:		
Size: A2	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 20/22

WATER PUMP MOTOR CONTROL



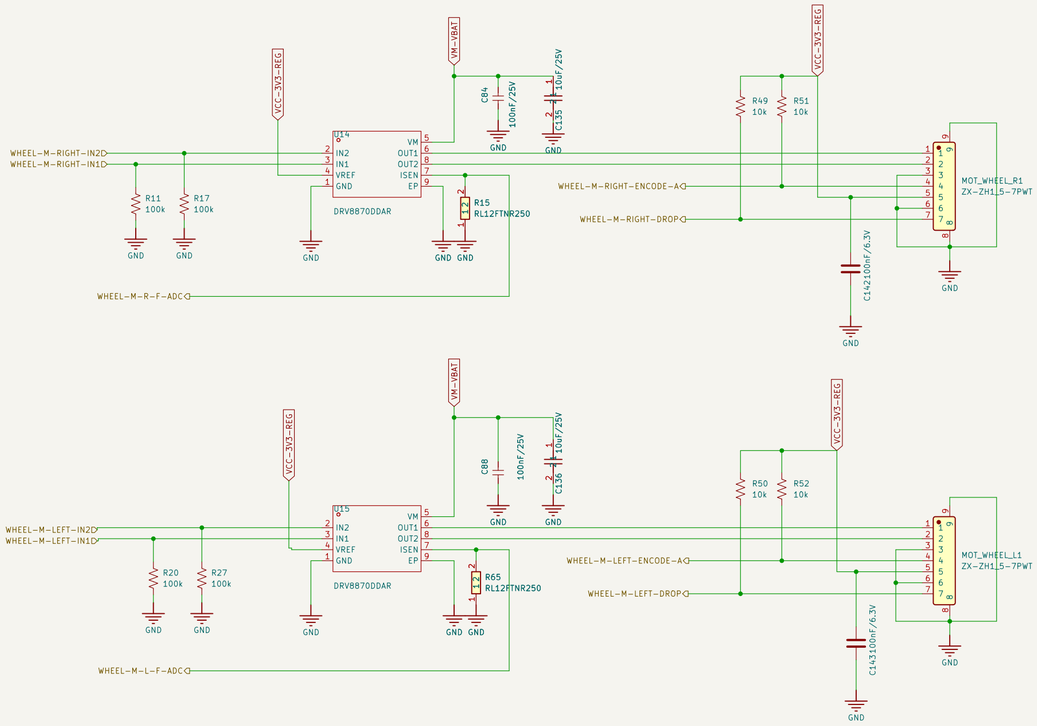
Sheet: /WATER-PUMP/	
File: WATER-PUMP.kicad_sch	
Title:	
Size: A2	Date:
KiCad E.D.A. 10.0.4	**Rev:**
	Id: 21/22

MAIN BUSH MOTOR CONTROL



Sheet: /MAIN-BRUSH MOTOR/		
File: MAIN-BRUSH-MOTOR .kicad_sch		
Title:		
Size: A2	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 22/22

WHEEL MOTOR CONTROL

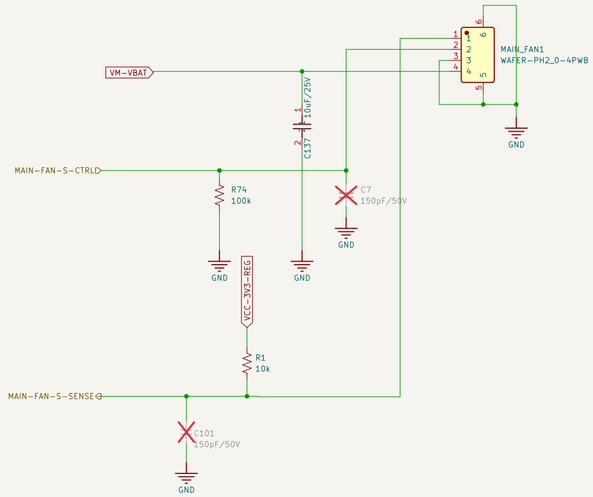


Sheet: /WHEEL MOTOR/
File: WHEEL-MOTORs .kicad_sch

Title:

Size: A2 | Date: | **Rev:**
KiCad E.D.A. 10.0.4 | Id: 23/22

MAIN FAN CONTROL

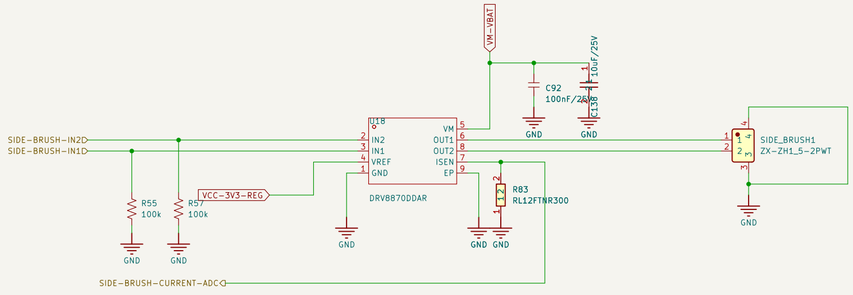


Sheet: /MAIN-FAN/
File: MAIN-FAN .kicad_sch

Title:

Size: A2	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 24/22

SIDE-BRUSH MOTOR CONTROL



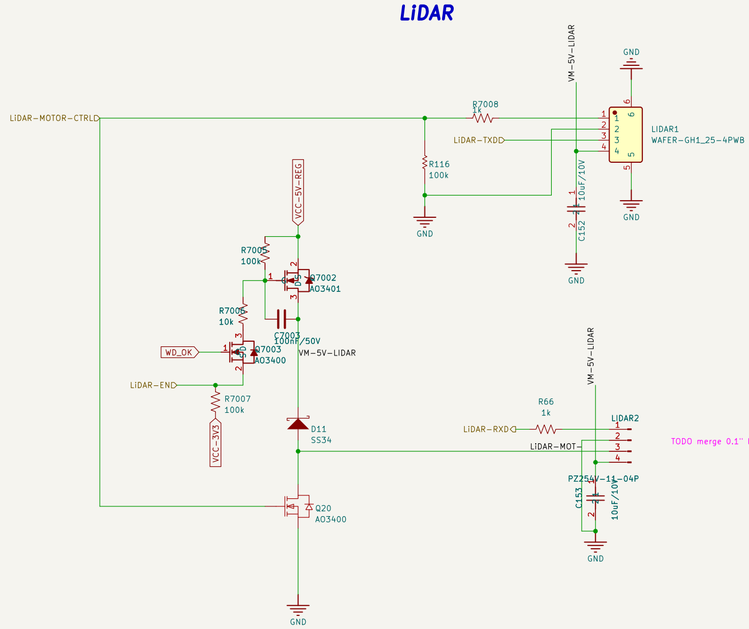
Sheet: /SIDE-BRUSH/
File: SIDE-BRUSH-MOTORs .kicad_sch

Title:

Size: A2	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 25/22

VM-5V-LIDAR powers the whole LiDAR (laser included) for any model. On only while WD_OK is high
AND LiDAR-EN is low (Q7003 source-switched AND). R7006/C7003: ~1.5 ms soft start, ~30 ms soft stop.
LIDAR2: 0.1in header for other LiDARs (pin 2 now GND).

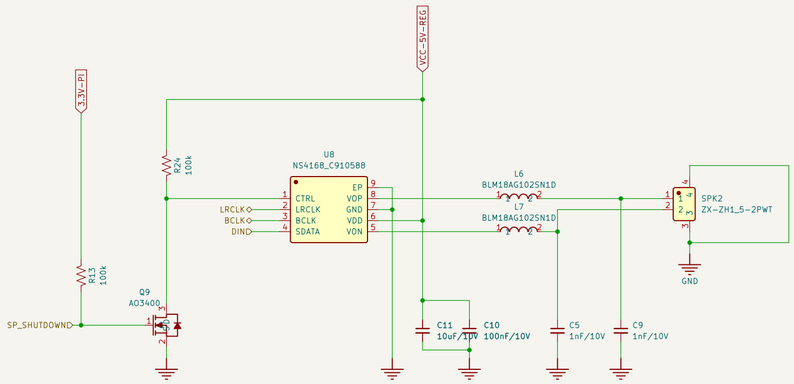
LiDAR



TODO merge 0.1" headers to cut BoM lines, costs

Sheet: /LiDAR/	
File: LiDAR.kicad_sch	
Title:	
Size: A2	Date:
KiCad E.D.A. 10.0.4	**Rev:** Id: 26/22

SPEAKER

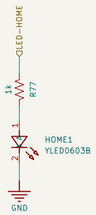


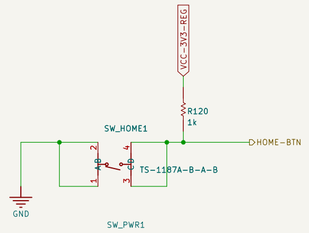
Sheet: /SPEAKER/
File: SPEAKER.kicad_sch

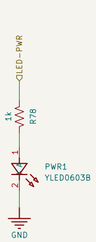
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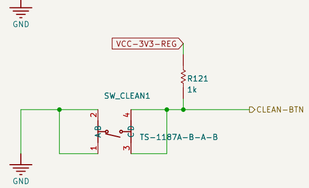
Size: A2	Date:	Rev:
KiCad E.D.A. 10.0.4		Id: 27/22

LEDs AND BUTTONs



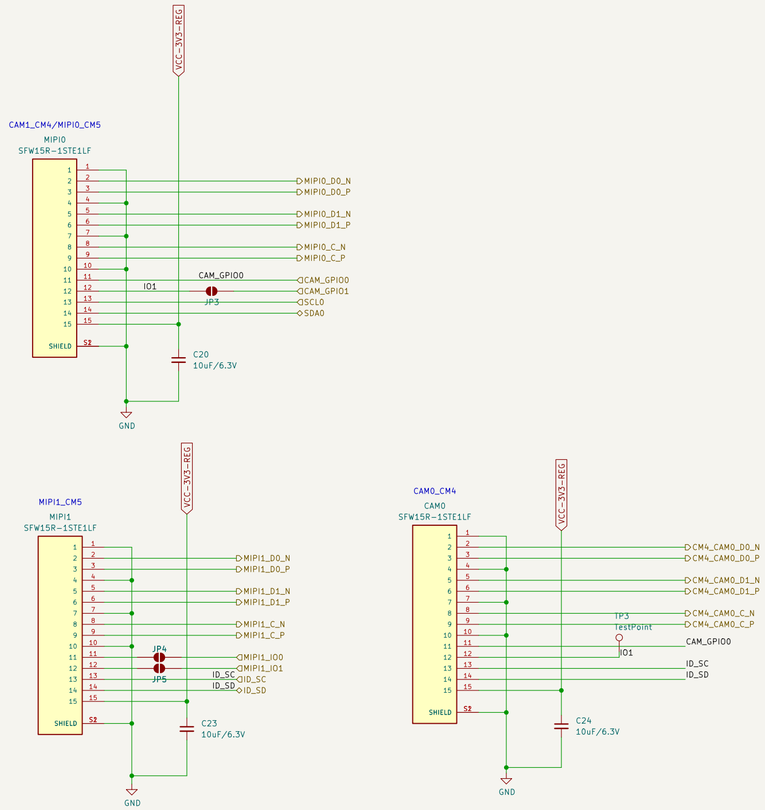






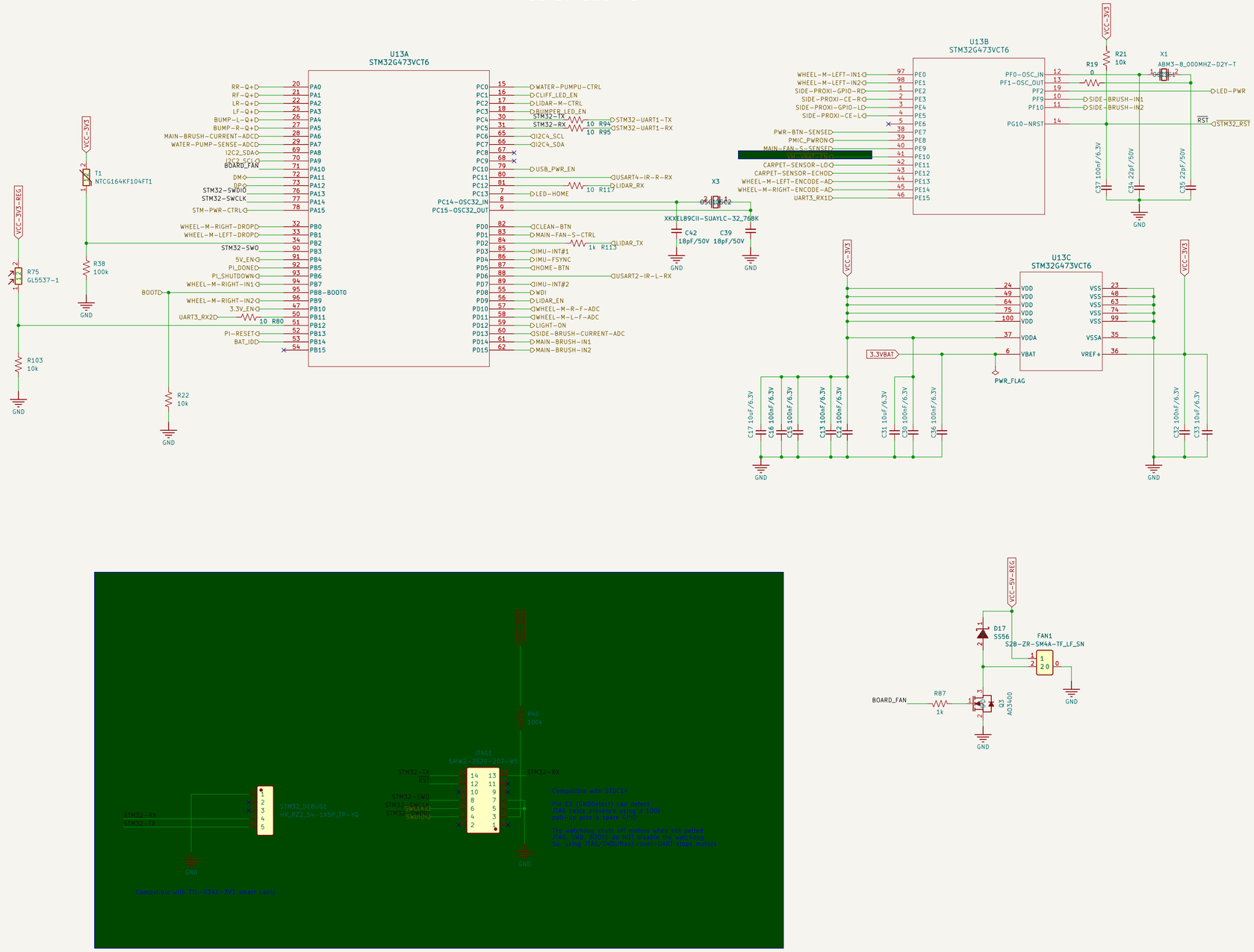
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KiCad E.D.A. 10.0.4		Id: 28/22

MIPI Cameras
2 lanes / camera



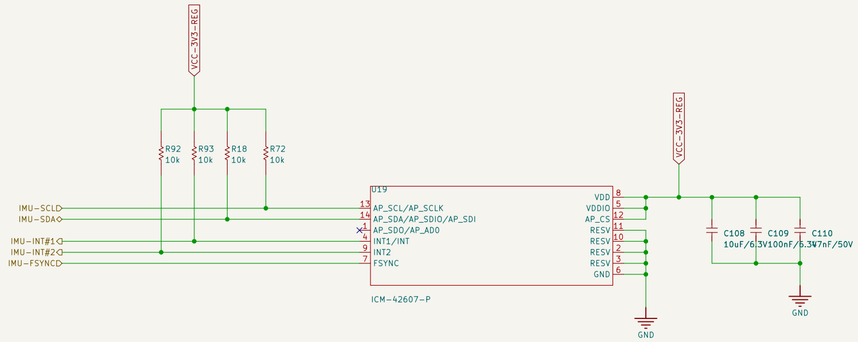
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MCU STM32G473



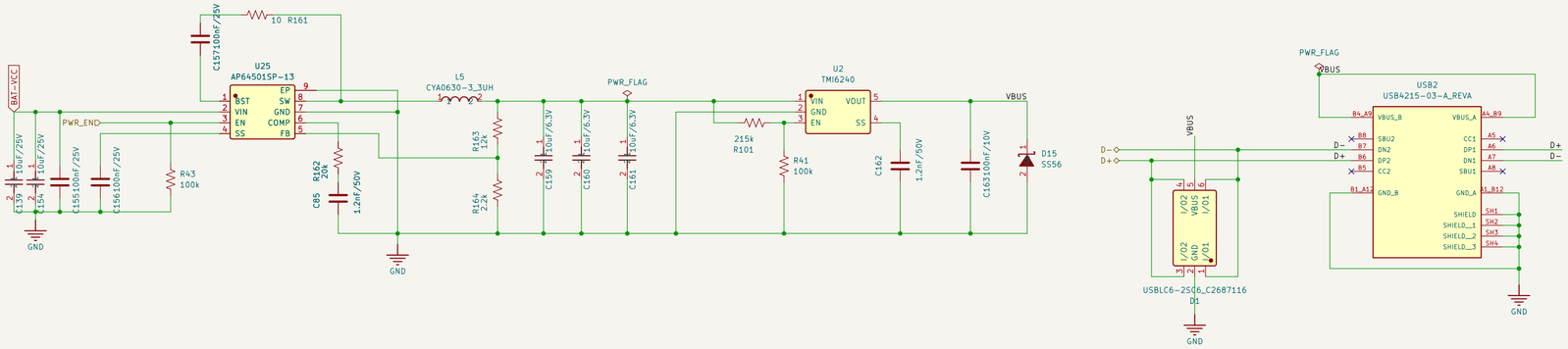
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IMU-ICM-42607-P

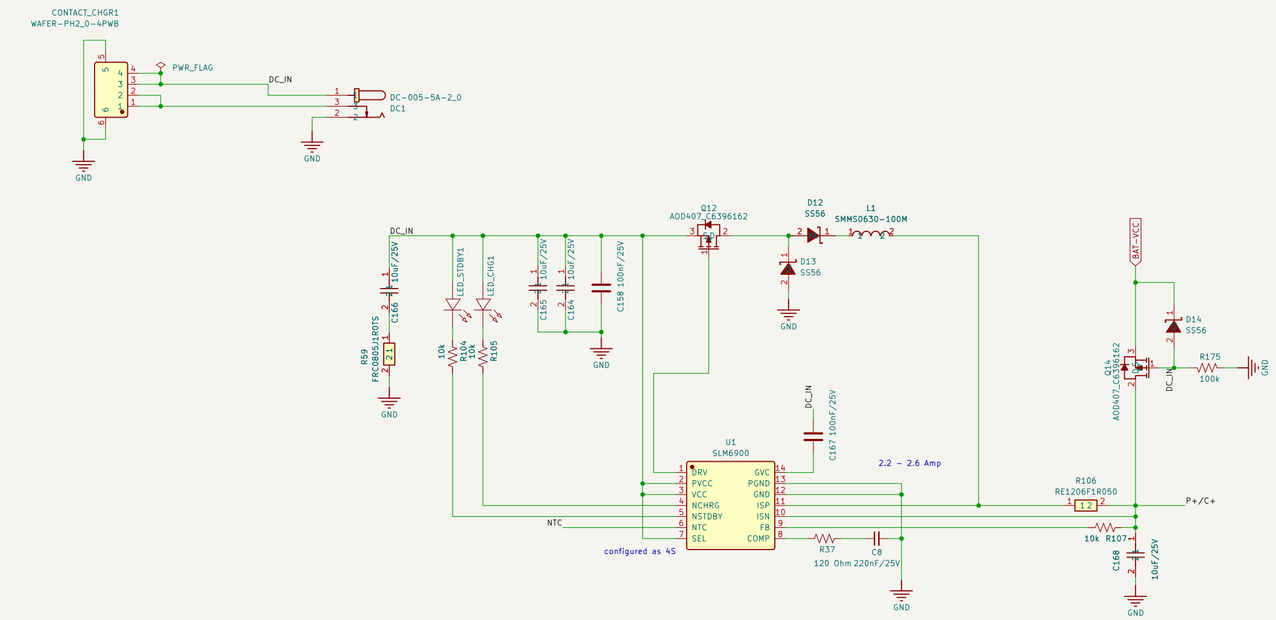


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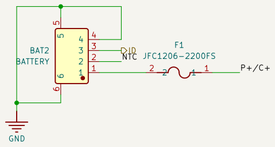
USB-C



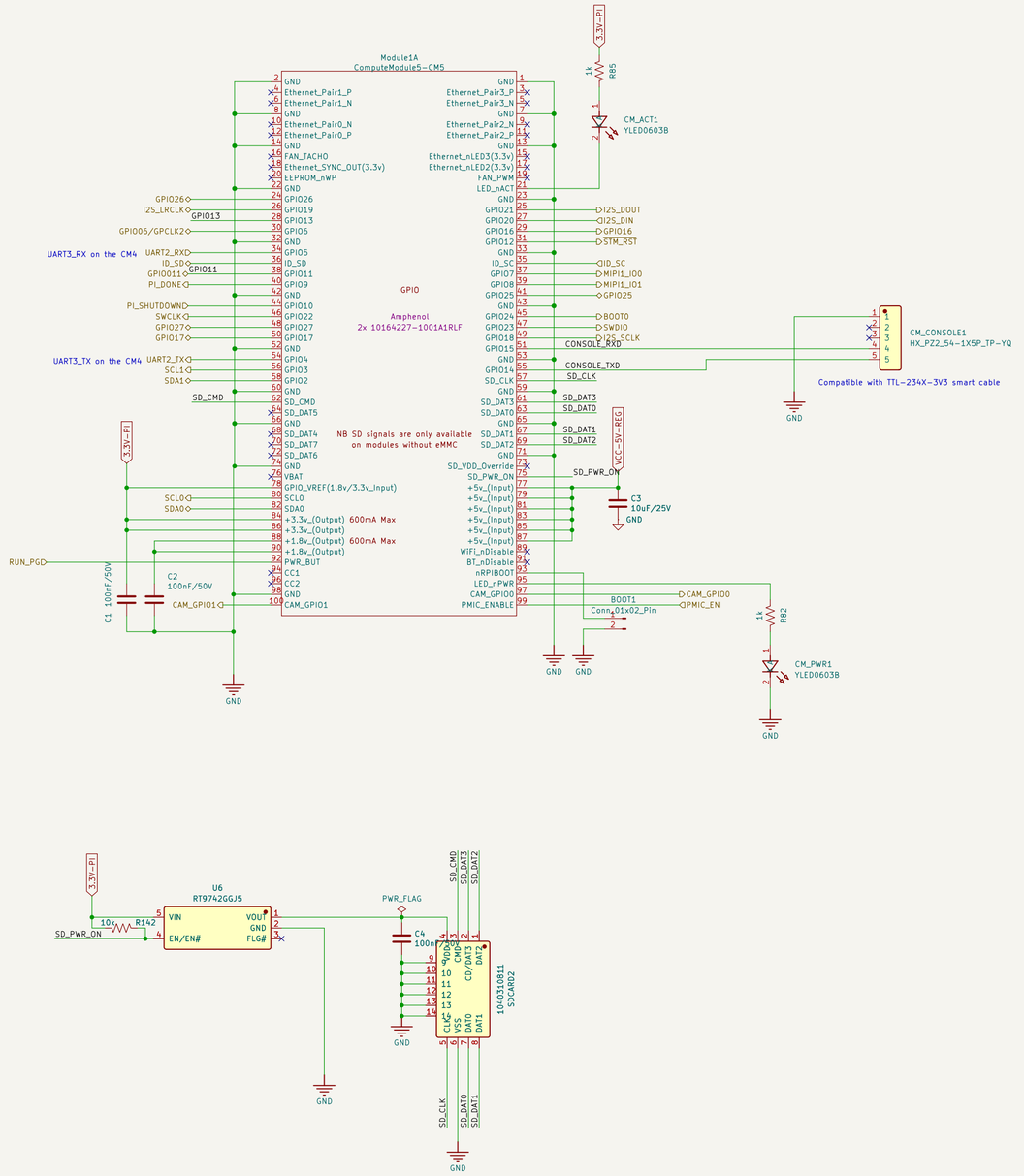
BATTERY CHARGER



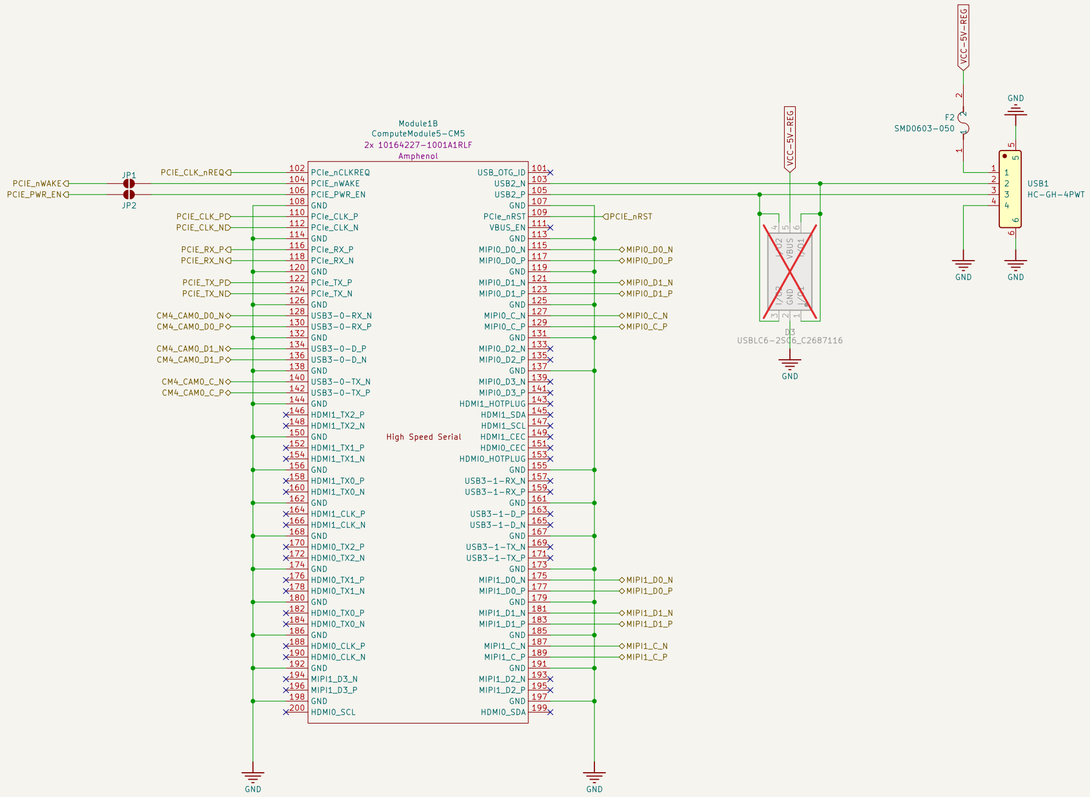
BATTERY CONNECTOR



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